Trinidad Borrell

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Summary

I'm a physicist turned neuroscientist and deep learning researcher, currently a PhD candidate at the Paris Brain Institute and Forschungszentrum Jülich. My trajectory runs from the mathematical foundations of particle physics, through the study of human and machine robustness to adversarial attacks, and into deep learning for clinical brain signals. This has meant working hands-on with transformer-based foundation models and graph variational autoencoders on EEG data from patients with disorders of consciousness. Additionally, I am interested in AI safety, particularly mechanistic interpretability, and was selected as a MARS V Fellow, where I lead a project on characterizing the persona space of large language models.

Experience

AI Safety Fellow **Jun 2026 – Today**
MARS V Programme *Oxford, England*

- Selected for a highly competitive AI safety research mentorship programme (5% acceptance rate) run by the Cambridge AI Safety Hub, collaborating in a team to produce publishable work. Conducting mechanistic interpretability research on persona manifolds in large language models. Mentored by Thomas Jiralerspong.

Doctoral Candidate **Oct 2025 – Today**
Paris Brain Institute | Forschungszentrum Jülich *Paris, France | Jülich, Germany*

- Benchmarked seven time-series and EEG foundation models (TOTEM, LaBram, NeuroLM, CbraMod, EEGPT, BIOT and SleepFM) for classification of patients with disorders of consciousness, establishing the first systematic evaluation framework for this clinical population; Decoder-only based and EEG pretrained models consistently outperformed general-purpose time-series models across the majority of metrics. *github.com/doc-benchmark*
- Designed and trained a Graph Autoencoder over EEG functional connectivity matrices using GraphSAGE based message passing and subject-level data splitting, demonstrating that the latent space captures meaningful brain connectivity patterns across patient states. *github.com/gnn-connectivity*
- Built a hybrid CNN and GNN pipeline for context-aware cancer cell classification on the NuCLS histopathology dataset, encoding spatial relationships between nuclei as graph edges and developing a class-balanced data augmentation strategy to mitigate severe label imbalance across 8 cell types; achieved 0.65 AUC, competitive with pretrained CNN baselines *github.com/gnn-cancer-cell*

AI Engineer **Apr 2025 – Oct 2025**
Paris Brain Institute *Paris, France*

- Developed preprocessing and marker computation pipelines for multi-subject clinical EEG data. Containerized the full pipeline with Docker to enable reproducible execution across research environments.

AI Researcher **Feb 2024 – Apr 2025**
Consciousness, Culture and Complexity lab *Buenos Aires, Argentina*

- Investigated the differential impact of adversarial perturbations on human visual perception and CNNs (EfficientNet, ResNet, Inception V3); presented as first-author at Khipu Latin American Conference in AI 2025 *github.com/adv-attacks*
- Developed a LLM and RAG pipeline over transcribed mind-wandering patient recordings, enabling structured extraction and analysis of spontaneous thought patterns from clinical interview data.

Research Intern **Mar 2023 – Mar 2024**
Lambda Lab *Buenos Aires, Argentina*

- Assessed coded aperture techniques for X-ray detection through Monte Carlo-based computational analysis *github.com/coded-aperture*

Education

Sorbonne Université **Sept 2025 – Today**
PhD Candidate in Neuroscience *Paris, France*

University of Buenos Aires **Mar 2019 – Mar 2025**
Licentiate in Physics (Equivalent to Bachelor and Master on Science) *Buenos Aires, Argentina*
GPA (in 10-point scale): 9.4

Teaching & Mentoring Experience

AI Safety Research Manager **Oct 2026 – Today**
AISAR *Remote*

- Supervising fellows' research projects from methodology to execution, connecting their work to the AI safety literature and field priorities, and advising on career paths and applications to research programs.

AI Safety Research Mentor **Aug 2026 – Today**
SPAR *Remote*

- Mentoring SPAR fellows on whether curvature-aware methods reveal that concept geometry is shared across different language models.

Technical AI Safety Facilitator **May 2026 – Today**
BlueDot *Remote*

- Guiding cohorts of emerging AI safety researchers through core concepts in alignment, mechanistic interpretability, evaluations, and AI control.

Data Science Teacher **Jul 2021 – May 2022**
CoderHouse *Buenos Aires, Argentina*

- Delivered live online instruction in data science, teaching Python, data analysis, visualization, and machine learning through practical, project-based learning.

Papers

G. Marraffini, G. Mahuas, **T. Borrell**, V. Shevchenko, D. Wassermann. The Variance Brain Foundation Models Forgot: Third-Order Statistics Predict Cognition Where Billion-Parameter Models Fail. arXiv:2606.04010.
N. Waehner, E. Tagliazucchi, J D. Sitt, D. Wassermann, G. Marraffini, and **T. Borrell**.The Platonic Brain: Representational Convergence in EEG Foundation Models. NeurIPS 2026 Foundation Models for the Brain and Body Workshop.
M. Carro, D. Mester, F. Nieto, O. Stanchi, G. Bergman, G. Marraffini, **T. Borrell** et al. Humans Can Correct Their Judgements Through Debate, but Weaker AI Models Can Not. NeurIPS 2026 Main Conference.
T. Borrell, S. Dillmann, K. Duraphe, F. Eris, A. Khederlarian, A. Kumar, G. Marraffini, MJ. Smith, S. Sourav, R. Di Tella, and JF. Wu. The Platonic Universe: Do Foundation Models See the Same Sky?. ICML 2026 Mechanistic Interpretability Workshop.
T. Borrell, L. Belloli, G. Marraffini, K R Patil, J D. Sitt, and F. Raimondo. What Have EEG Foundation Models Actually Learned About the Conscious Brain?. *Preprint*

Grants

BlueDot Rapid Grant. ICML 2026 Full Travel Grant.
Iliad Intensive 2026. Full Travel Grant and Scholarship.
BlueDot Rapid Grant. Compute Grant.
ML4Good Bootcamp 2026. Full Travel Grant and Scholarship.
Winter School AI for Sciences 2026 Full Travel Grant and Scholarship.
Paris Brain Institute America Doctoral Grant. 3-year PhD Grant.
Khipu Latin American Meeting in Artificial Intelligence 2025 Full Travel Grant.
Brain, Minds and Machines 2024 Summer School Full Travel Grant and Scholarship.

Presentations and Posters

T. Borrell, L. Belloli, G. Marraffini, K R Patil, J D. Sitt, and F. Raimondo. (June, 2026). *A Comprehensive Evaluation Framework for EEG Foundation Models in Disorders of Consciousness.* Poster presented at OHBM 2026. Bordeaux, France.
T. Borrell, E. Tagliazucchi. (March, 2025). *Subtle or obvious? Confidence reports indicate the success of adversarial attacks on humans.* Poster presented at Khipu Latin American Meeting in Artificial Intelligence 2025. Santiago, Chile.
T. Borrell, E. Tagliazucchi. (November, 2024).*Human and machine vulnerability to adversarial perturbations.* Poster presented at the Scientific Symposium on AI and Applications, led by the University of Saint Andrews. Buenos Aires, Argentina.
T. Borrell, E. Tagliazucchi. (September, 2024). *Human and machine vulnerability to adversarial perturbations.* Poster presented at the 2024 Annual Reunion of the Argentinian Physics Asociation. San Luis, Argentina.
T. Borrell, E. Sprejer, A. G. Magnoni, D. Rodrigues, Q. Pears Stefano. (September, 2023). *Impossible Photographs.* Poster presented at the 2023 Annual Reunion of the Argentinian Physics Asociation. Buenos Aires, Argentina.

Skills

AI/ML: Python, PyTorch, HuggingFace, FastAPI, Scikit-learn, Claude Code, Time Series Foundation Models, Fine-tuning, Evaluation, Optimization, LLMs, Tansformers, RAG, LangGraph, LangChain Prompt Engineering, CNNs, Synthetic Data Generation, Agentic Engineering, GenAI.
Data: Pandas, NumPy, SQL, MongoDB.
Infrastructure: Docker, Git, Github, Docker, HTCondor.
Visualization: Plotly, Seaborn, Matplotlib.
Languages: Spanish (Native), English (TOEFL iBT: 106), French (DELF B1: 78).